

Diffuse optimum certificates and integer recovery

Research continuation, 7 October 2026. We prove a finite theorem producing an optimal fractional cover whose individual weights decrease as the rank grows. We then combine it with an explicitly imported classical rounding theorem to recover integer covers on a noninteger interval. The argument does not assume small pair intersections. It does not resolve the entire question stated as Kahn's Conjecture 5.5 in 1994.

1. Statements and conventions

Let H be a finite simple intersecting hypergraph with m nonempty edges, each of size at most the integer r , and maximum vertex degree at most the integer D . Throughout $r \geq 2$ and $D \geq 3$. Its integer vertex cover number is $\tau(H)$; its fractional vertex cover number is $\tau^*(H)$. A fractional cover x assigns nonnegative vertex weights with total at least one on each edge. Isolated vertices are discarded. For convenience set

$$\Delta = (D - 1)m - D(D - 2)r - D, \quad K_D = D(D - 2). \quad (1)$$

Theorem 1 (finite diffuse optimum). If $\Delta > 0$, then $\tau^*(H) = m/D$. There is an optimal fractional cover x such that

$$0 \leq x_v \leq \frac{1}{\lceil \Delta/K_D \rceil} \leq K_D/\Delta \quad \text{for every vertex } v. \quad (2)$$

Every optimal cover has weight zero at vertices of degree less than D , and total weight exactly one on each edge. The existence of a cover in (2) is proved by a finite repair and a minimax argument, not numerical optimization. No pair- or triple-intersection restriction is needed.

Define

$$\chi_D = D - 1 - \frac{2}{3D(D - 2) + 2(D - 1)}. \quad (3)$$

For example, $\chi_3 = 24/13$ and $\chi_4 = 44/15$.

Theorem 2 (integer recovery without small pair intersections). Fix D . Suppose H_j are as above, $r_j \rightarrow \infty$, $m_j/r_j \rightarrow c \in (\chi_D, D - 1]$, and

$$\lambda_j := \max_{E, F, G \text{ distinct}} |E \cap F \cap G| = o(r_j). \quad (4)$$

The maximum is zero if fewer than three edges occur. Then

$$\tau^*(H_j) = m_j/D \quad \text{eventually,} \quad \tau(H_j) = m_j/D + o(r_j), \quad \tau(H_j)/\tau^*(H_j) \rightarrow 1. \quad (5)$$

For $D = 3$ the interval is $(24/13, 2]$. For $D = 4$ it is $(44/15, 3]$. The bound on vertex degree is a real hypothesis. The proof uses Kayll's published bounded-edge-size rounding theorem and Edmonds's graph matching polytope theorem, both stated in Section 5.

Corollary 3 (fractional extremizers in this interval). Let $\sigma_D = D - 1 - 1/(D - 1)$. Fix $c \in (\chi_D, D - 1)$. For a simple intersecting r -uniform sequence with $m/r \rightarrow c$, $r \rightarrow \infty$, and (4), suppose

$$\tau^*/r \rightarrow c/D. \quad (6)$$

There is no initial bound on vertex degrees in this corollary. Then $\tau/r \rightarrow c/D$. Equivalently, the previously proved characterization by deleting $o(r)$ edges to leave maximum degree at most D now also implies integer recovery under small triple intersections on this interval. We specify the precise extraction input in Section 7.

The narrower range in (3) is intentional. We do not claim the entire strict fractional ramp $(\sigma_D, D - 1)$, a general integer frontier, or a solution of Ryser’s conjecture. The present status of Kahn 5.5 is not settled by our limited literature screen. No first-result claim is made.

2. The finite fractional input

We use the following already proved repository theorem. For any simple intersecting family of m nonempty edges of rank at most $r \geq 2$ and every integer $k \geq 2$,

$$\tau^* \leq \max \left\{ \frac{(k - 1)r + 1}{k}, \frac{m}{k + 1} \right\}. \quad (7)$$

Its complete proof appears in the attributed predecessor *The sharp fractional cover frontier at every finite edge/rank ratio*, commit c5255aaf66f9e894c6cbb0e2d1eb47c738c23765, Sections 2–3. It uses a maximum dual edge weight, assigns every other edge to exactly one common vertex, and sums a signed bin inequality. The finite bound is independent of any design existence, rounding theorem or OpenAI result. Its actual finite incidence implication has a partial Lean formalization there.

In our setting, uniform dual edge weights $1/D$ are feasible, because a vertex belongs to at most D edges. Thus $\tau^* \geq m/D$. Apply (7) with $k=D-1$. The strict inequality $\Delta > 0$ says exactly that

$$\frac{(D - 2)r + 1}{D - 1} < \frac{m}{D}.$$

Consequently $\tau^* = m/D$.

For any optimal cover x , summing its edge loads gives

$$m \leq \sum_E \sum_{v \in E} x_v = \sum_v d(v)x_v \leq D \sum_v x_v = m. \quad (8)$$

All quantities are finite and every summand is nonnegative. Equality throughout implies edge loads one, and $(D - d(v))x_v = 0$ for each vertex. In particular positive weights are supported on degree- D vertices. This proves the asserted support and load conditions without using complementary slackness as an unstated assumption.

The optimal-cover set is nonempty and compact. For completeness, any nonnegative cover weight may be truncated to one without violating a cover constraint. Minimize the sum over the resulting finite box. Once the optimal value is m/D , (8) implies every active coordinate is at most one; the optimal set is a closed subset of the box. Hence the largest coordinate has a minimum on it.

3. Removing degree-D vertices and repairing intersections

Lemma 4 (repair). Let R be any set of vertices, all of degree D , and let L be an integer such that each original edge contains at most L vertices of R . There is a simple intersecting hypergraph H_R , on the same m edge labels, whose maximum vertex degree is at most D and rank at most $r + L$. All retained original vertices have their original incidence sets. Every new vertex has degree strictly less than D .

Proof. Delete R . For each removed vertex v , partition its D incident edge labels into three nonempty groups $G_{v,1}, G_{v,2}, G_{v,3}$. This is possible since $D \geq 3$. Create three fresh vertices $z_{v,1}, z_{v,2}, z_{v,3}$, where $z_{v,i}$ belongs precisely to the incident labels outside $G_{v,i}$. Every incident label receives exactly two of them. The new vertex $z_{v,i}$ has degree $D - |G_{v,i}|$, between two and $D-1$. All new vertices are distinct from one another and from all original vertices.

An edge containing t removed vertices loses t vertices and receives $2t$ new ones. Its size is therefore

$$|E| + t \leq r + L. \quad (9)$$

Every original edge pair still meets: a retained common vertex remains, or at a removed common vertex its two labels together exclude at most two group indices, so they share at least one of the three new vertices. Nonempty edges remain nonempty, and old vertex degrees are unchanged.

Simplicity also survives. For distinct original edges E, F , choose an original vertex belonging to one of them and not the other. If it is retained it still distinguishes them. If it was removed, its replacement places two fresh vertices on the incident label and none on the other label. Thus the repaired edges remain distinct. This proves the lemma.

The three-complement repair is stronger than replacing a removed vertex by all pairwise intersections: its rank cost is only one per removed incidence, for every $D \geq 3$. No degree- D replacement vertex is introduced.

If

$$\Delta > K_D L, \quad (10)$$

then (7), applied to H_R with rank parameter $r + L$, again gives $\tau^*(H_R) = m/D$. Formula (8) for H_R implies every new vertex of degree below D has optimal weight zero. Therefore an optimal cover of H_R uses only retained degree- D original vertices, and defines an optimal cover of H avoiding R . This is a finite resilience statement; R need not be small in total size, only in its incidence count on each edge.

4. Proof of the diffuse optimum theorem

Choose an optimal cover x minimizing $M = \max_v x_v$. We have $M > 0$ because $m > 0$. Let R be the nonempty set of vertices where $x_v = M$. They all have degree D by (8). At an edge, its load one shows

$$|E \cap R| M \leq 1.$$

Put $L = \max_E |E \cap R|$, so $L \leq 1/M$. Suppose that $M > K_D/\Delta$. Then $K_D L \leq K_D/M < \Delta$. Lemma 4 and (10) provide another optimal cover x' which is zero on R .

For every coordinate outside R we have $x_v < M$. There are finitely many coordinates, so for sufficiently small positive ε ,

$$(1 - \varepsilon)x_v + \varepsilon x'_v < M \quad (v \notin R).$$

On R the same convex combination equals $(1 - \varepsilon)M < M$. The combination remains an optimal cover, contradicting minimality of M . Thus $M \leq K_D/\Delta$. For the sharper rounded version put $N = \lceil \Delta/K_D \rceil \geq 1$. If $M > 1/N$, then $L \leq 1/M < N$, so the integer L satisfies $L \leq N - 1 < \Delta/K_D$. The same repair and convex-combination contradiction applies. Thus $M \leq 1/N$, completing Theorem 1. The finite proof covers nonuniform families as well as uniform ones. It is not a fully formalized linear-program argument; the exact formal scope is recorded in the accompanying formal README.

5. Imported rounding and matching-polytope facts

Pass to the incidence dual J . Its m points are the edges of H ; each original vertex v determines a block B_v of its incident edge labels. The blocks have size at most D . A vertex cover of H is precisely a block cover of J . Distinct original vertices may give identical blocks. When needed merge such blocks, summing their fractional weights; any chosen merged block has an original representative, so integer cover cost does not increase. All codegrees below count original block occurrences.

For a block-weight vector t , let $\alpha_3(t) = \max_{p,q,s \text{ distinct}} \sum_{B \supset \{p,q,s\}} t(B)$. For S a point set define the local restriction, on subsets A of S , by

$$t|_S(A) = \sum_{B \cap S = A} t(B) \quad (|A| \geq 2), \quad t|_S(A) = 0 \quad (|A| < 2). \quad (11)$$

Let $MP(S)$ be the convex hull of incidence vectors of matchings of these subsets (each matching consists of pairwise disjoint subsets of size at least two). Let $b(t)$ be the largest integer such that every set S of at most that size has $t|_S \in MP(S)$. Ignoring singleton coordinates in (11) is part of the Kayll theorem's definition, not a change of the cover constraints of J .

Input KQ (Kayll 1995, Theorem 2.1). For fixed D , a D -bounded hypergraph J and a fractional block cover t satisfy

$$\rho(J) \leq (1 + o(1)) \sum_B t(B) \quad (12)$$

whenever $\alpha_3(t) \rightarrow 0$ and $b(t) \rightarrow \infty$. Here ρ denotes minimum integer block cover. The precise readable source is P. M. Kayll, *Asymptotically Good Covers in Hypergraphs: Extended Abstract of the Dissertation*, DIMACS Technical Report 95-55 (1995), printed p.5, Theorem 2.1; definitions appear immediately above it. This theorem generalizes Kahn 1994 Conjecture 5.6. The later Kahn–Kayll paper has DOI 10.1006/jcta.1997.2761. We read the extended abstract's full primary statement, not that later paper's inaccessible full text. The rounding theorem is imported, not reproved or formalized in this package.

Input E (Edmonds's graph matching polytope). Nonnegative graph-edge weights w lie in the convex hull of graph matchings if and only if each vertex load is at most one and for every odd vertex set T , $\sum_{e \in T} w_e \leq (|T| - 1)/2$. Consequently, every such vector with maximum weighted degree at most $2/3$ belongs to that polytope: for odd $|T| \geq 3$, its internal mass is at most $|T|/3 \leq (|T| - 1)/2$. The one-point constraints are vacuous. Source: J. Edmonds, *Maximum matching and a polyhedron with 0,1-vertices*, J. Res. Nat. Bur. Standards B 69 (1965), 125–130. This standard characterization is another stated input; the elementary degree-to-odd-set implication is proved here.

6. Local control and proof of integer recovery

The argument also gives a finite local certificate. If t is a fractional tiling on the incidence dual, $t(B) \leq M$, and the maximum triple codegree is λ , then for any integer $h \geq 2$ with

$$\epsilon = \binom{h}{3} M \lambda < 1, \quad M \{(D - 1)r - m + h\} \leq \frac{2}{3}(1 - \epsilon), \quad (13a)$$

we have $b(t) \geq h$. This uses only the stated pair coverage, degree/rank bounds, and Input E. The following proof establishes this finite criterion as well as its asymptotic application.

Take the cover x from Theorem 1 and write $t(B_v) = x_v$. By (8) it is a fractional tiling of J , meaning each point has load exactly one, with $\sum_B t(B) = m/D$. Every positive block has size D , and $t(B) \leq M$, where $M = K_D/\Delta$.

Here is a useful counting inequality independent of these weights. Fix a set S of h points and p in S . Pair coverage implies

$$\sum_{q \notin S} d(p, q) \geq m - h, \quad \sum_{q \neq p} d(p, q) = \sum_{B \ni p} (|B| - 1) \leq (D - 1)r.$$

Subtracting gives the **local pair budget**

$$\sum_{q \in S \setminus \{p\}} d(p, q) \leq (D - 1)r - m + h. \quad (13)$$

Each block at p which meets S at least twice contributes at least one to the left. Hence the total nontrivial restriction load at p is at most

$$M\{(D - 1)r - m + h\}. \quad (14)$$

This uses the actual incidence count, even if a block meets S more than twice. It is not a claim about an arbitrary optimal cover.

Along the sequence in Theorem 2 set $\delta = D - 1 - c$. We have

$$rM \longrightarrow C = \frac{D(D - 2)}{1 - (D - 1)\delta}, \quad C\delta < 2/3, \quad (15)$$

because (3) is exactly $\delta < 2/\{3D(D - 2) + 2(D - 1)\}$. The denominator is strictly positive. The original family also satisfies $m - 1 \leq (D - 1)r$ by pair coverage, explaining $c \leq D - 1$.

Moreover $\alpha_3(t) \leq M\lambda \rightarrow 0$. Choose an integer h_j tending to infinity slowly enough that

$$Mh_j \rightarrow 0, \quad h_j^3 M\lambda \rightarrow 0. \quad (16)$$

For example the integer part of $(r/(\lambda + 1))^{1/4}$ works eventually: $M = O(1/r)$, $\lambda/r \rightarrow 0$, and h_j/r tends to zero. For every S of size $h \leq h_j$, the total mass $\eta = \sum_{|A| \geq 3} t|_S(A)$ is bounded uniformly by

$$\eta \leq \binom{h}{3} M\lambda = o(1), \quad (17)$$

since every block counted there contains a triple of S . Multiple counting only increases the upper bound.

Let w be the pair coordinates of $t|_S$. For $h \geq 2$ its weighted degree is at most the bound in (14). Equations (15)–(17) imply, uniformly over all such S , that the maximum weighted degree of $w/(1 - \eta)$ is strictly less than $2/3$ eventually. Input E supplies a convex combination of graph matchings realizing $w/(1 - \eta)$. Multiply this distribution by $1 - \eta$ and, for each $|A| \geq 3$, add probability $t|_S(A)$ on the matching containing the single subset A . The probabilities total one and the resulting matching-incidence expectation is precisely $t|_S$. For $h < 2$ the restriction is zero and the empty matching suffices. Thus $b(t) \geq h_j \rightarrow \infty$.

This step checks all local higher-subset coordinates: it does not replace their hypergraph matching polytope by a graph polytope without accounting for them. It also explains why a degree bound strictly below $2/3$ is useful rather than a mere vertex-load bound of one.

Input KQ now gives $\tau(H) = \rho(J) \leq (1 + o(1))m/D$. The lower bound $m/D = \tau^*(H) \leq \tau(H)$ proves (5), since $m = \Theta(r)$ and $c > 0$. This completes the proof of Theorem 2.

7. Extending to fractional extremizers

The strict-ramp extraction proved at commit c5255a in the predecessor states: for fixed $c \in (\sigma_D, D-1]$, intersecting r -uniform families with $m/r \rightarrow c$ and $\tau^*/r \rightarrow c/D$ admit deletion of $o(r)$ edges to leave maximum vertex degree at most D . Its proof selects edges of dual weight greater than $1/(D+1)$, using the quantitative peak estimate in its Theorem 4. This is a previously proved repository input, not a new contribution here. The endpoint $c = D - 1$ is also covered by the earlier integer-ratio extraction.

For Corollary 3 choose this core. Its edge count is $m - o(r)$, its rank is at most r , and its triple intersections are still $o(r)$. Apply Theorem 2. Each deleted nonempty edge can be covered with one extra vertex, so $\tau(H) \leq m/D + o(r)$. The lower bound (6) proves the corollary. No control on pair intersections of the original family or the core is introduced. The same reasoning covers $c = D - 1$ using the stated endpoint extraction, although integer recovery at that endpoint was already proved in the predecessor.

8. Scope, examples and limitations

Small pair intersections genuinely need not follow from our hypotheses. For example take the incidence dual of an affine-line design on $\mathbb{F}_3^N$, whose edges have rank $r_0 = (3^N - 1)/2$, whose vertices have degree three, and whose pair intersections are one. For two chosen edges, add t fresh vertices belonging exactly to that pair. Pad every other edge with t fresh private vertices. The result is simple, intersecting and $(r_0 + t)$ -uniform, with maximum degree three, maximum triple intersection one, and maximum pair intersection $t + 1$. If $t/r_0 \rightarrow \theta \in (0, 1/12)$, then

$$m/r \rightarrow 2/(1 + \theta) > 24/13, \max |E \cap F|/r \rightarrow \theta/(1 + \theta) > 0.$$

It therefore falls within Theorem 2 and outside the small-pair input. This example is illustrative; its integer recovery also follows from its underlying design and is not an independent breakthrough.

The finite certificate (2) is not asserted optimal in its constant. The threshold χ_D comes from that constant and the sufficient $2/3$ graph condition; neither is asserted sharp. For $D = 3$ the remainder of the strict ramp, $(3/2, 24/13]$, is not settled by this proof. Nor does the proof settle families without the degree/core condition, or with triple intersections of order r . Improving the diffuse constant, or using sharper local odd-set information instead of a degree-only test, are concrete further questions.

The complete written argument uses finite fractional-cover duality, compactness, the predecessor finite theorem, Edmonds's theorem, and Kayll's theorem exactly as disclosed. The rational checker verifies repair incidences, feasible primal/dual certificates and local matching distributions on fixed examples; it is not a proof of the infinite rounding input. The Lean companion proves selected finite inequalities, not the minimax existence argument, Edmonds/Kayll inputs or the full asymptotic theorem. Literature comparison remains incomplete.